

Lab 03: Automated Liquid-Receiving Tank Ladder Diagram Implementation

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Environment: OpenPLC Editor (IEC 61131-3)

Executive Summary

This report provides a description of the design, implementation, testing, and safety verifications of the automated liquid-receiving tank PLC control system, which was developed for the ICDFA Lab 03 application. The program provides an automatic fill-cycle function within a prescribed range of 2.0" ft" to 8.0" ft" , while also providing discrete high/low process alarms at 9.0" ft" and 1.0" ft" , and a high-level safety interlock (Estop_Cmd) to prevent physical overfill events.

1. System Variable Configuration Table

All variables were configured in the OpenPLC Editor environment using standard IEC 61131-3 data types and setpoint definitions matching the official ICDFA Variable Plan:

Variable Name	Class	Data Type	Setpoint / Default	Description & Operational Purpose
TankFillAuto_Cmd	Local	BOOL	FALSE	Internal automatic fill latch/command (Bistable SET/RESET state).
TankValveOpen	Output	BOOL	FALSE	Discrete output command energizing the physical fill valve solenoid.
Estop_Cmd	Input	BOOL	FALSE	Emergency-stop discrete input; TRUE forces immediate auto-fill termination.
TankLevelHi_Alarm	Output	BOOL	FALSE	Discrete high-level alarm indicator coil.
TankLevelLo_Alarm	Output	BOOL	FALSE	Discrete low-level alarm indicator coil.
TankLevel_LIT	Input	REAL	Analog Input	Measured continuous tank liquid level in feet (0.0 to 10.0 ft).
TankEmpty_SP	Local	REAL	2.0	Automatic fill start setpoint threshold (2.0 ft).
TankFull_SP	Local	REAL	8.0	Automatic fill stop setpoint threshold (8.0 ft).

Variable Name	Class	Data Type	Setpoint / Default	Description & Operational Purpose
TankLoAlarm_SP	Local	REAL	1.0	Low-level process alarm setpoint threshold (1.0 ft).
TankHiAlarm_SP	Local	REAL	9.0	High-level process alarm setpoint threshold (9.0 ft).

SCREENSHOT 1: Variable Table Setup

(OpenPLC Variable Configuration Table showing defined BOOL/REAL tags and initial values)

ICDFA-PLC-Engineering-v1.0 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Project

- Unnamed
 - TankFill
 - Res0

Config0.Res0.instance0

- TankFillAuto_Cmd (BOOL)
- TankValveOpen (BOOL)
- Esto_Cmd (BOOL)
- TankLevelHi_Alarm (BOOL)
- TankLevelLo_Alarm (BOOL)
- TankLevel_LIT (REAL)

Search

TankFill

Description: Class Filter: All

#	Name	Class	Type	Location	Initial Value	Option	Documentation
1	TankFillAuto_Cmd	Local	BOOL		FALSE		
2	TankValveOpen	Local	BOOL		FALSE		
3	Esto_Cmd	Local	BOOL		FALSE		
4	TankLevelHi_Alarm	Local	BOOL		FALSE		
5	TankLevelLo_Alarm	Local	BOOL		FALSE		
6	TankLevel_LIT	Local	REAL		0.0		
7	TankEmpty_SP	Local	REAL		2.0		
8	TankFull_SP	Local	REAL		8.0		
9	TankLoAlarm_SP	Local	REAL		1.0		
10	TankHiAlarm_SP	Local	REAL		9.0		
11	TankFill_VAL	Local	BOOL		FALSE		

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- Time
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- Bitwise
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- Compariso

2. Ladder Logic Architecture & Rung Breakdown

Rung 1: Valve Actuation & High-Level Safety Interlock

- Logic:** TankFillAuto_Cmd contact in series with a negated / TankLevelHi_Alarm / contact drives the TankValveOpen output coil.

- **Function:** Permits valve actuation during automatic filling only if no high-level alarm condition exists.

Rung 2: Automatic Start Latch (SET Condition)

- **Logic:** An execution-controlled LE block evaluates whether $\text{TankLevel_LIT} \leq \text{TankEmpty_SP}$ (2.0 ft).
- **Function:** When the tank level drops to or below 2.0 ft, the output energizes TankFillAuto_Cmd (S) to initiate auto-filling.

Rung 3: Automatic Stop & Emergency Interlock (RESET Condition)

- **Logic:** Parallel OR branches evaluating $\text{TankLevel_LIT} \geq \text{TankFull_SP}$ (8.0 ft), $\text{Estop_Cmd} == \text{TRUE}$, OR $\text{TankLevelHi_Alarm} == \text{TRUE}$ feed into TankFillAuto_Cmd (R).
- **Function:** Resets the automatic fill command state if the tank reaches full capacity, the E-stop is triggered, or a high-level alarm trips.

Rung 4: High-Level Alarm Logic

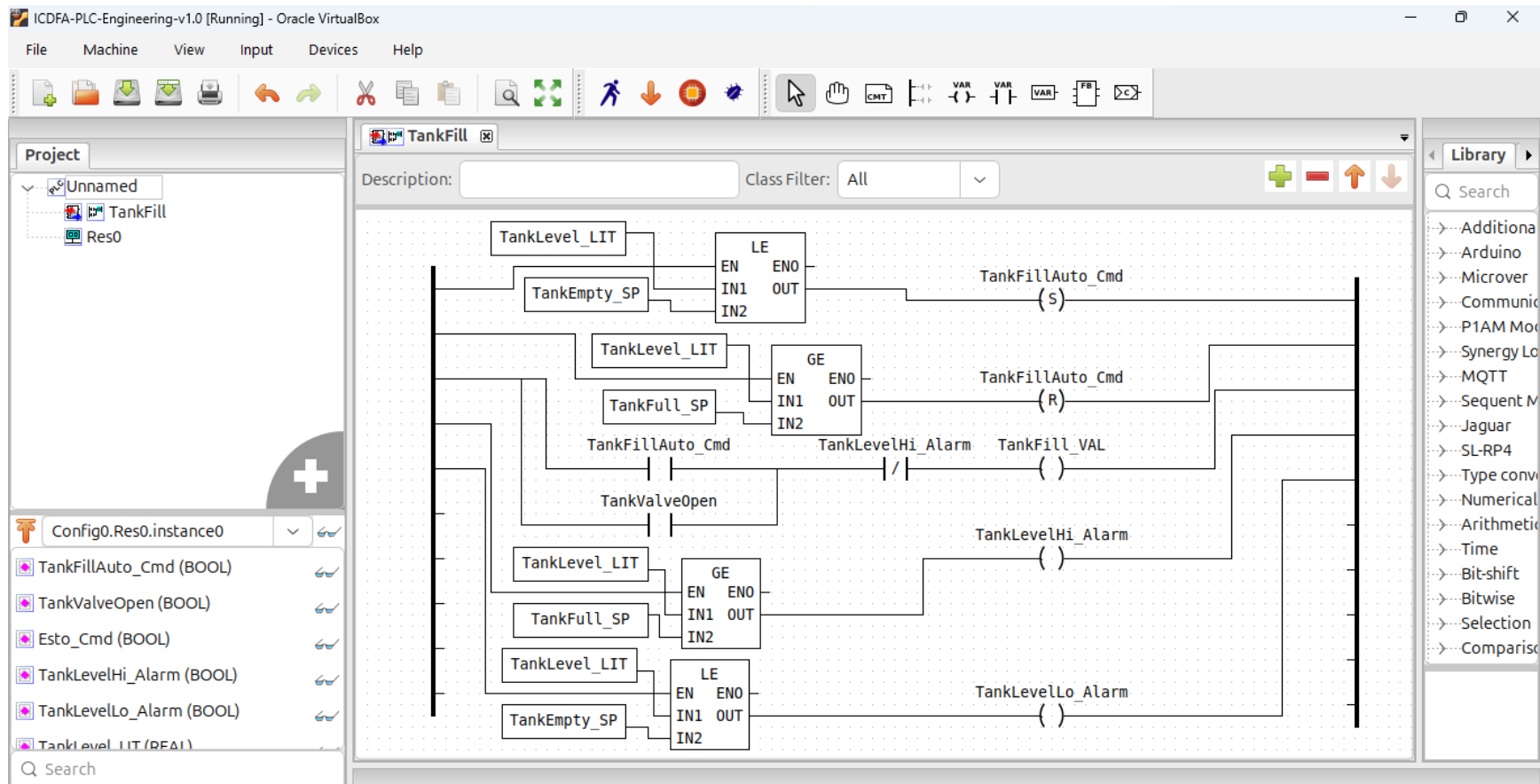
- **Logic:** An execution-controlled GE block compares TankLevel_LIT against TankHiAlarm_SP (9.0 ft).
- **Function:** Energizes TankLevelHi_Alarm () when liquid level reaches or exceeds 9.0 ft.

Rung 5: Low-Level Alarm Logic

- **Logic:** An execution-controlled LE block compares TankLevel_LIT against TankLoAlarm_SP (1.0 ft).
- **Function:** Energizes TankLevelLo_Alarm () when liquid level drops to or below 1.0 ft.

[SCREENSHOT 2: Complete 5-Rung Ladder Program]

(Complete OpenPLC Ladder Diagram showing auto-fill logic, interlocks, and alarm rungs)



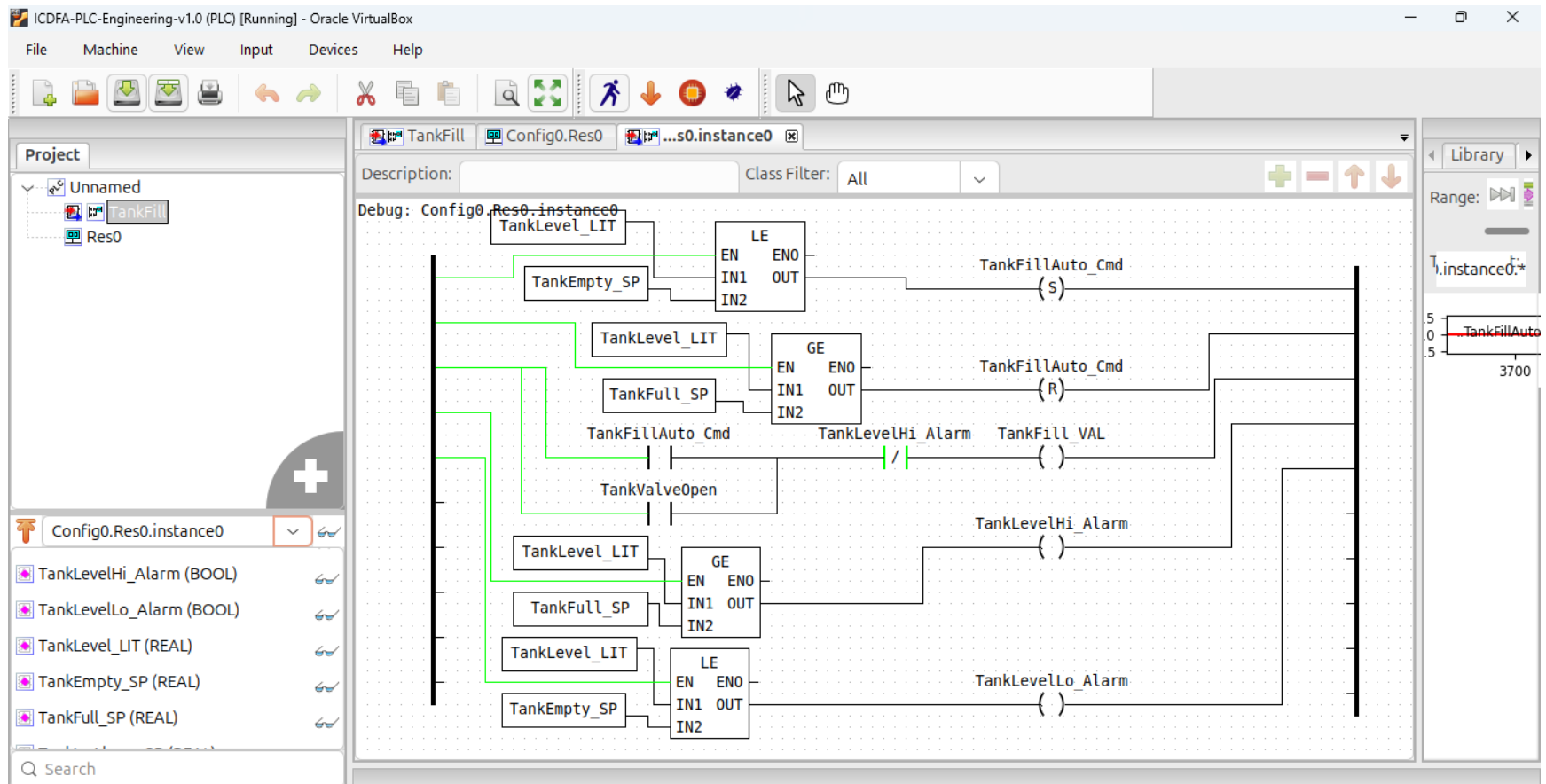
3. Six-Scenario Logic Test Table

Scenario #	Test Scenario Description	Key Inputs / Conditions	Expected TankFillAuto_Cmd	Expected TankValveOpen	Active Alarm Outputs
1	Empty Tank Start	TankLevel_LIT = 0.5 ft, Estop_Cmd = FALSE	TRUE (Set)	TRUE	TankLevelLo_Alarm (TRUE)
2	Normal Filling Mode	TankLevel_LIT = 5.0 ft, Estop_Cmd = FALSE	TRUE (Latched)	TRUE	None (FALSE)
3	Full Tank Cutoff	TankLevel_LIT = 8.2 ft, Estop_Cmd = FALSE	FALSE (Reset)	FALSE	None (FALSE)
4	High Alarm Condition	TankLevel_LIT = 9.5 ft, Estop_Cmd = FALSE	FALSE	FALSE (Interlocked)	TankLevelHi_Alarm (TRUE)
5	Low Alarm Condition	TankLevel_LIT = 0.8 ft, Estop_Cmd = FALSE	TRUE (Set)	TRUE	TankLevelLo_Alarm (TRUE)

Scenario #	Test Scenario Description	Key Inputs / Conditions	Expected TankFillAuto_Cmd	Expected TankValveOpen	Active Alarm Outputs
6	Emergency Stop Override	TankLevel_LIT = 1.5 ft, Estop_Cmd = TRUE	FALSE (Reset)	FALSE (Forced Off)	TankLevelLo_Alarm (TRUE)

[SCREENSHOT 3: Simulation Debug Window]

(OpenPLC Instance Debugger showing active green power flow during simulation verification)



4. Safety Override Explanation

In industrial automation, an Emergency Stop (Estop_Cmd) condition must always override normal automatic process logic to make sure that the deterministic hardware safety takes place. Automatic control process works under the setpoint values but in the case of malfunctioning of valves, level sensors, ruptured pipes, loss of communication, etc., the automatic feed of the process material to the tanks will continue just because the system is designed to work under certain logic.

The described Estop_Cmd signal wire is routed to the RESET coil of the TankFillAuto_Cmd and the safety interlock coils so that in the case of emergency, a human operator could stop filling the tanks and cut off the power supply to the necessary valves and other equipment in order to avoid spills, tank overflow, chemical hazards, and other dangerous situations for personnel and the environment.

5. Official ICDFA Knowledge Check Solutions

Question 1: Why are BOOL and REAL appropriate data types for this project?

Answer: BOOL (Boolean) represents discrete two-state physical signals and logical status conditions (TRUE/FALSE), such as valve commands (TankValveOpen), emergency stop states (Estop_Cmd), auto-fill commands (TankFillAuto_Cmd), and discrete alarm coils (TankLevelHi_Alarm, TankLevelLo_Alarm).

REAL (single-precision floating-point) represents continuous analog engineering variables that require precision decimal values, such as the measured liquid transmitter level (TankLevel_LIT) and operational setpoint thresholds (TankEmpty_SP = 2.0 ft, TankFull_SP = 8.0 ft, TankLoAlarm_SP = 1.0 ft, TankHiAlarm_SP = 9.0 ft).

Question 2: What is the purpose of a SET/RESET pair for TankFillAuto_Cmd?

Answer: A SET/RESET latch pair creates a bistable memory control loop that maintains the automatic fill state across program scan cycles without requiring continuous trigger conditions. The SET coil latches TankFillAuto_Cmd to TRUE when the liquid level drops to or below 2.0 ft (TankEmpty_SP). The state remains latched in memory while filling until TankLevel_LIT reaches 8.0 ft (TankFull_SP), an active E-stop (Estop_Cmd), or a high alarm state energizes the RESET coil. This prevents rapid valve chattering or rapid cycling around setpoint boundaries.

Question 3: What should happen if the tank reaches 9 ft while the fill valve remains commanded open?

Answer: When TankLevel_LIT reaches 9.0 ft (TankHiAlarm_SP), the high-level alarm comparison block evaluates TRUE, energizing the TankLevelHi_Alarm coil. Energizing TankLevelHi_Alarm triggers two immediate safety actions:

1. It energizes the RESET coil of TankFillAuto_Cmd, latching the automatic fill command to FALSE.

2. It opens the normally closed / TankLevelHi_Alarm / interlock contact in Rung 1, instantly de-energizing TankValveOpen (FALSE) to isolate the fill valve and prevent overfilling.

Question 4: Why should PLC logic be validated in simulation before use with physical equipment?

Answer: Validating PLC logic in a simulated environment allows engineers to detect race conditions, logic errors, interlocking failures, and uncommanded output states safely before deployment to operational technology (OT) hardware. Simulation eliminates the real-world risks of physical equipment damage, chemical/liquid spills, pump burnout, overpressurization, or personnel injury caused by flawed ladder program code.

6. Conclusion

The TankFill control logic implemented in OpenPLC Editor meets all technical, operational, and safety requirements specified in ICDFA Lab 03. Through structured simulation testing, the system demonstrated reliable auto-fill latching between 2.0 ft and 8.0 ft, accurate high/low alarm triggering at 9.0 ft and 1.0 ft, and fail-safe safety overrides via Estop_Cmd and high-level safety interlocks.